

Fundamentals of Photovoltaics (EE338)

Lecture 2: Basic of light and solar spectrum

Light

- The light that we see everyday is only a fraction of the total energy emitted by the sun incident on the earth. Sunlight is a form of "electromagnetic radiation" and the visible light that we see is a small subset of the electromagnetic spectrum shown in Figure 2.1.
- The electromagnetic spectrum describes light as a wave which has a particular wavelength.

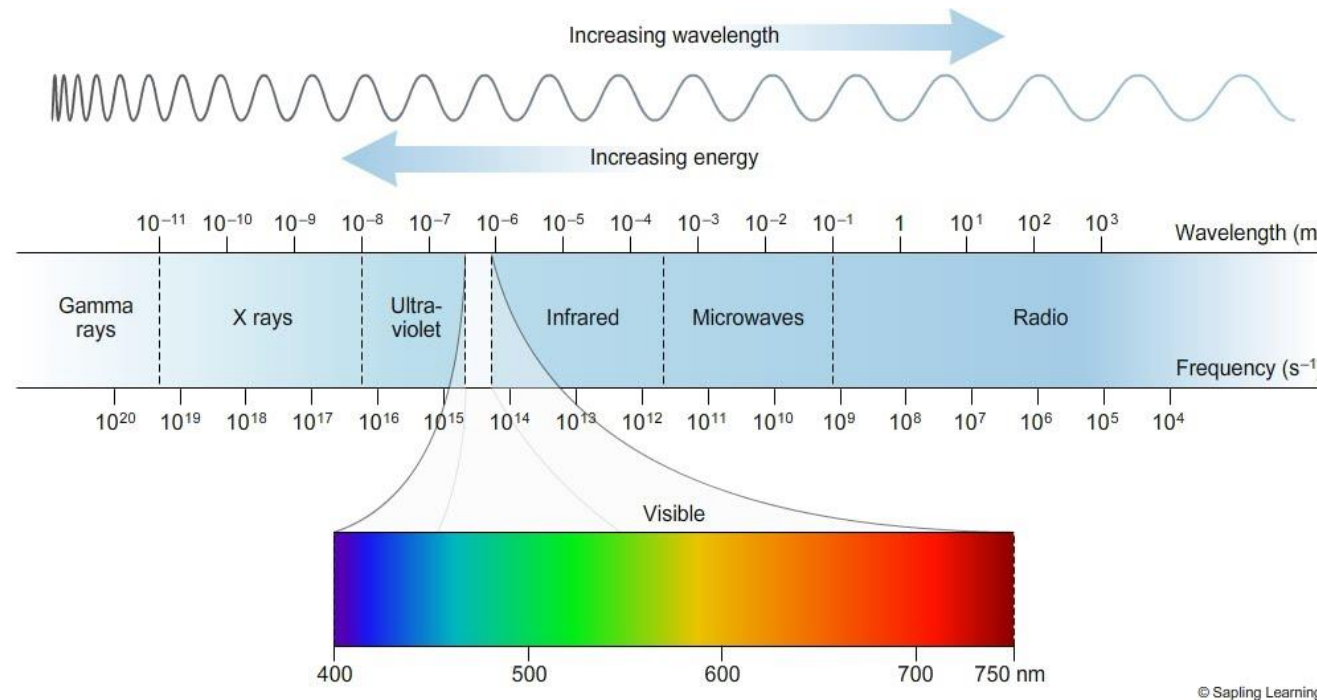


Figure 2.1. Electromagnetic wave spectrum

Evolution in Light Theory

- Newton thought of light as particles.
- However, in early 1800's when experiments by Thomas Young, François Arago, and Augustin Jean Fresnel showed interference effects in light beams, indicating that light is made of waves.
- So, light is an electromagnetic wave as shown in Figure 2.2, just like radio waves and microwaves with a wavelength, λ and electric field, E .

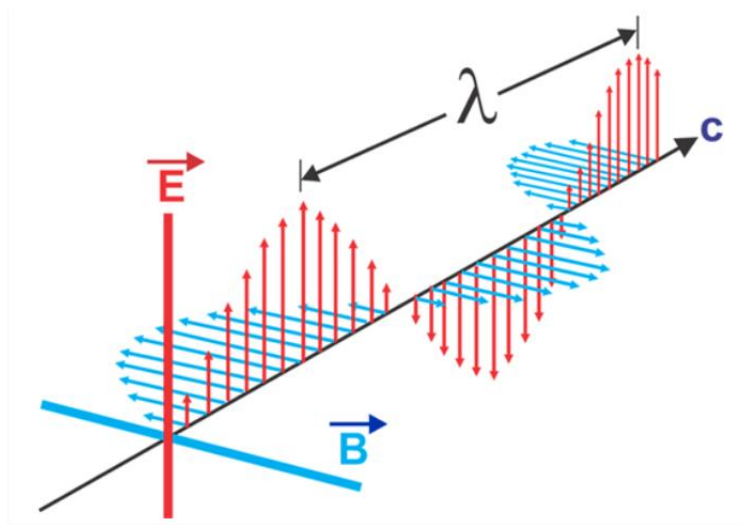
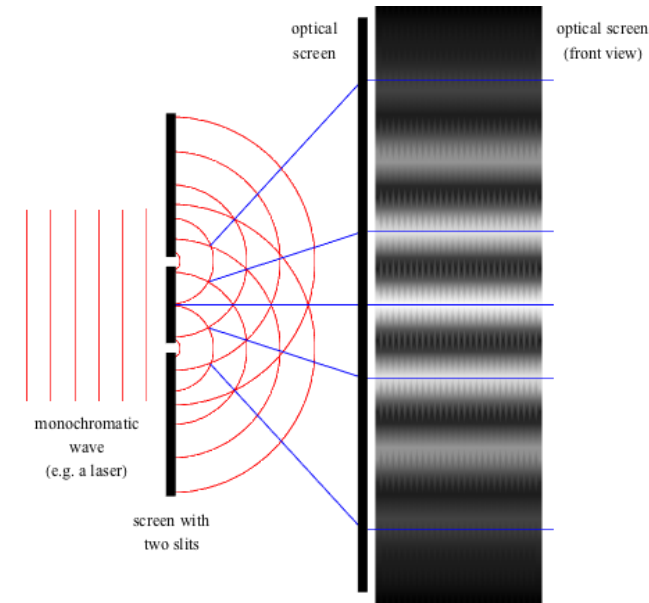


Figure 2.2. An electromagnetic wave



- Einstein noted that when the light of a certain colour hits a metal, all of the electrons that escape the metal have the same peak energy. The energy of the ejected electrons is determined by

$$K_{\max} = hf - \phi \quad (2.1)$$

(f is the frequency of incident photon, ϕ is the work function of metal and h is Planck constant)

- When the light intensity is increased, the number of electrons increases, not the energy of each electron.
- The energy in wave theory is E^2 , but the energy also equals $n_{\lambda} \times e_{\lambda}$, where is n_{λ} the number of photons and e_{λ} is the energy of each photon.
- So one can think of the electromagnetic wave intensity as getting smaller and smaller until one discovers that it is coming in discrete particles.
- Based on the photoelectric effect, Einstein said that light comes in small energy packets called quanta, which behave somewhat like particles.

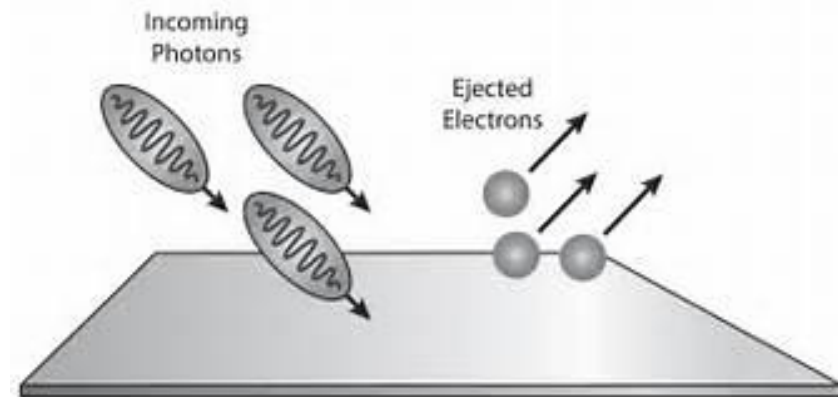
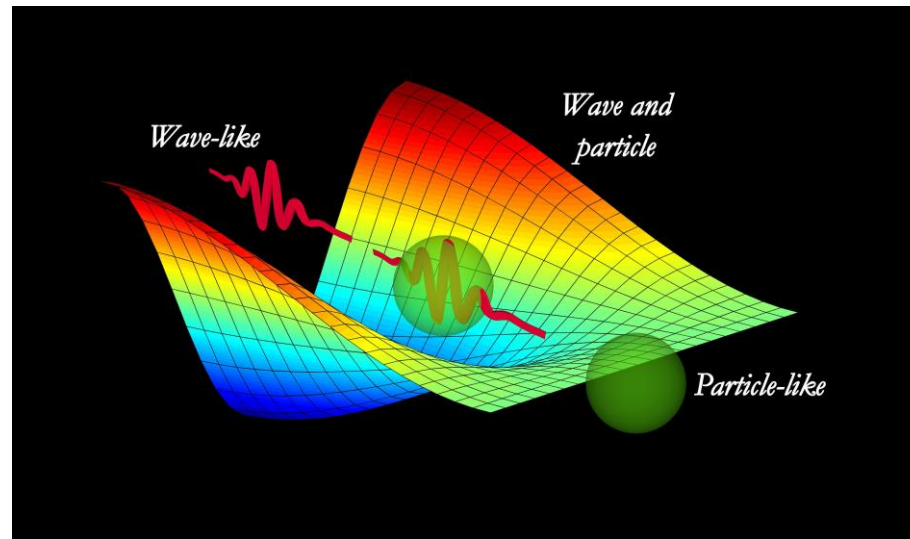


Figure 2.3. Illustration of photoelectric effect

- Today, quantum-mechanics explains both the observations of the wave nature and the particle nature of light.
- In quantum mechanics, a photon, like all other quantum-mechanical particles such as electrons, protons etc., is most accurately pictured as a "wave-packet".
- A wave packet is defined as a collection of waves which may interact in such a way that the wave-packet may either appear spatially localized or may alternately appear simply as a wave.
- In the cases where the wave-packet is spatially localized, it acts as a particle. Therefore, depending on the situation, a photon may appear as either a wave or as a particle and this concept is called "wave-particle duality".



Photon Energy

- The energy contained by a photon, E , is given by the equation:

$$E = \frac{hc}{\lambda} \quad (2.2)$$

where λ is the wavelength (in m), h is Planck's constant (6.626×10^{-34} J.s), c is the speed of light in vacuum (2.998×10^8 m/s)

The unit of E in SI is joule (J). However, the energy contained in a photon energy is rather small and can be better expressed by the unit electron-volt (eV).

1 eV is equivalent to 1.6×10^{-19} J.

By expressing the equation for photon energy in term of eV, a simplified equation which relates the energy and wavelength of a photon can be written as:

$$E(eV) = \frac{1.24}{\lambda (\mu m)} \quad (2.3)$$

- From the equation 2.3, it follows that photons with a short wavelength (ultra-violet and blue light) have a high energy and those with a long wavelength (red and infrared light), have a low energy.

High energy photon for blue light. 

Lower energy photon for red light. 

Low energy photon for infrared light.
Should be invisible! 

- Since the frequency ν (in Hz), wavelength λ and speed of light c , are related by $c = \lambda \cdot \nu$, the equation 2.2 also can be expressed as:

$$E = h \nu \quad (2.4)$$

- Quiz: What is the energy of a photon with wavelength of $0.6 \mu\text{m}$?

Ans: 2 eV

Light having a photon energy E_{ph} equal or higher than the bandgap energy E_g of a semiconductor is absorbed.

$$E_{ph} \geq E_g$$

$$\frac{1.24}{\lambda (\mu m)} \geq E_g (eV) \quad (2.5)$$

For example, the silicon has a bandgap of 1.1eV. Therefore, in a silicon solar cell, only the fraction of solar spectrum having wavelength shorter than 1.1 μm will be absorbed. The solar spectrum with wavelength above 1.1 μm will be lost. This is illustrated in Figure 2.4.

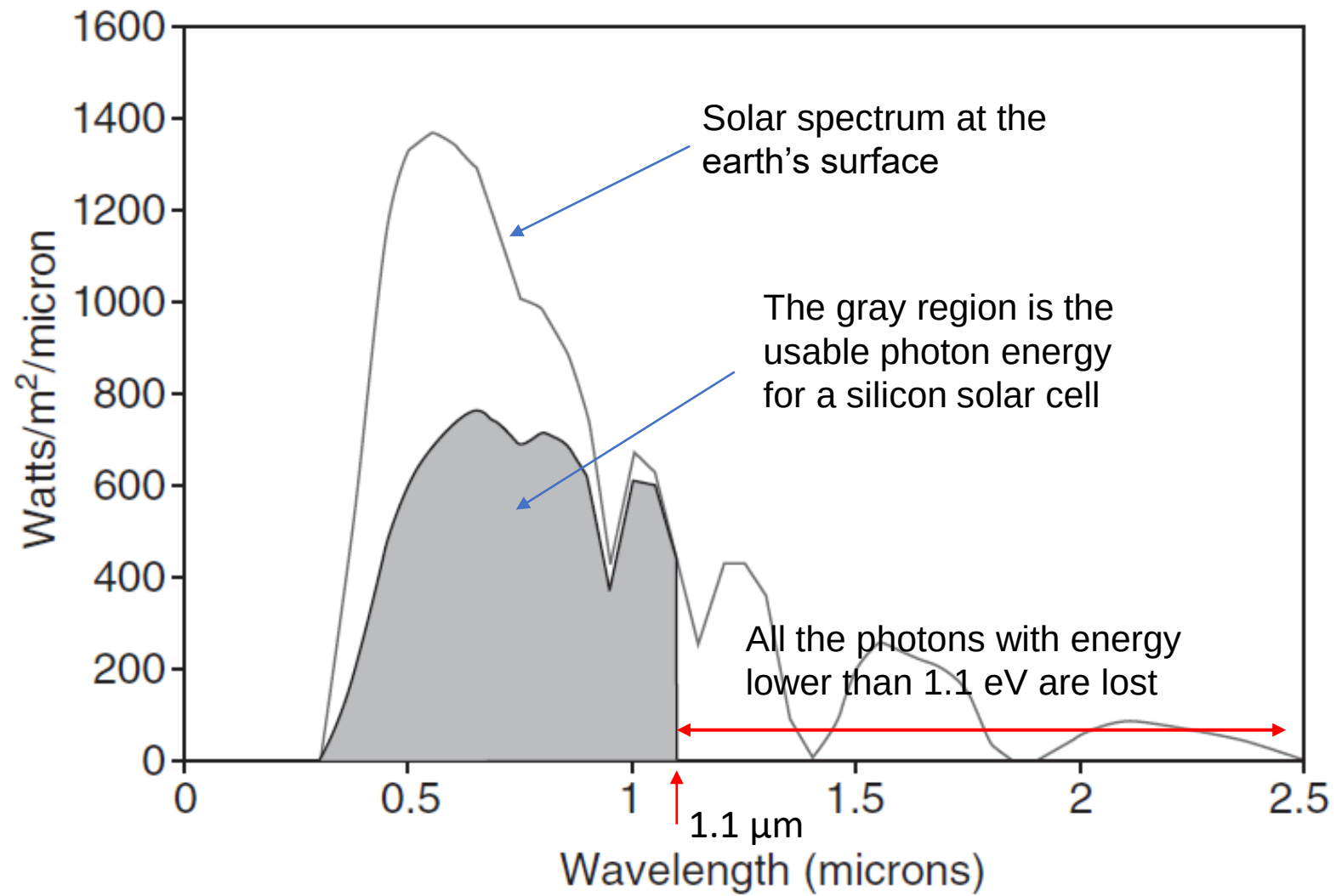


Figure 2.4

- For the photons with energies large enough to be absorbed (left side of the figure), each photon absorbed must create an electron that can move to generate. The yield here should be one electron per photon, but the actual number is less and is called the quantum efficiency (QE). (Note: Therefore, the grey area in the left is not identical to the solar spectrum.) QE of 90% occur in pure single crystal materials. QEs in non-single-crystal materials are lower.
- The photons absorbed are counted for current of a solar cell.
- While an absorbed photon has a higher energy than the bandgap of the absorbing material, it can only create an electron moving through a voltage of no more than the bandgap energy. For example, since silicon has a bandgap of 1.1 eV, a 2-eV photon loses at least $2.0 - 1.1 = 0.9$ eV.
- Hence, although a low bandgap material can absorb a wider solar spectrum (i.e., higher current), photons lose more energy (i.e., lower voltage).
- This leads to a way to make higher-efficiency solar cell using multiple material (multi-junction solar cell). Two stacked materials can absorb the lower-energy photons in one material generating a lower voltage and higher-energy photons in a second material generating a higher voltage. This results in **more photons utilization without losing so much photon energy**.

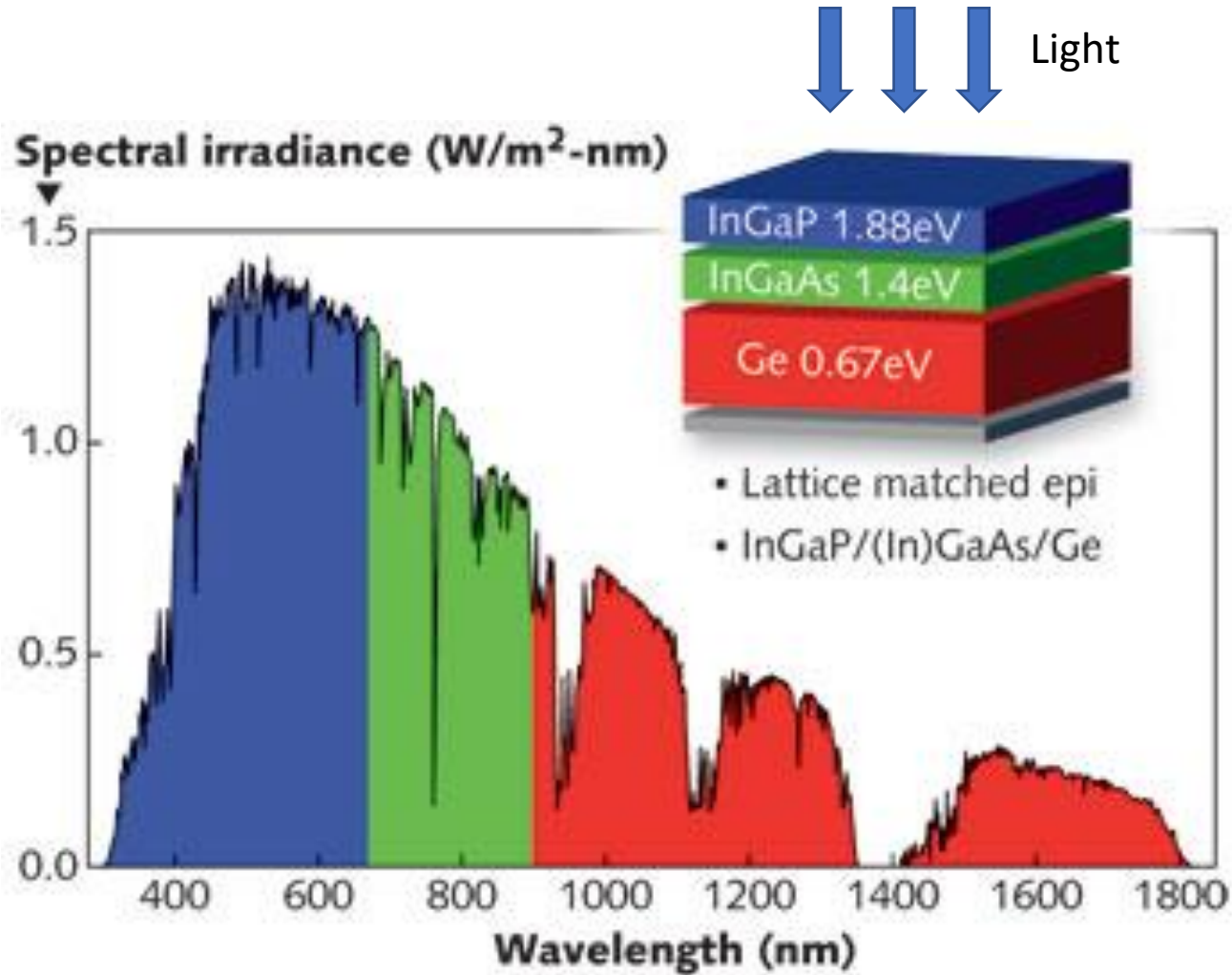


Figure 2.5

The structure and absorption of a conventional three-junction cell (Ga/InGaAs/InGaP).

The top InGaP layer absorbs photons with energies of at least 1.88 eV (~ 660 nm), transmitting lower-energy photons to the InGaAs with a 1.4 eV (~ 886 nm) bandgap. Wavelengths longer than 900 nm are transmitted to the 0.7 eV Ge layer, which absorbs out to 1800 nm, where sunlight reaching the ground drops to near zero.

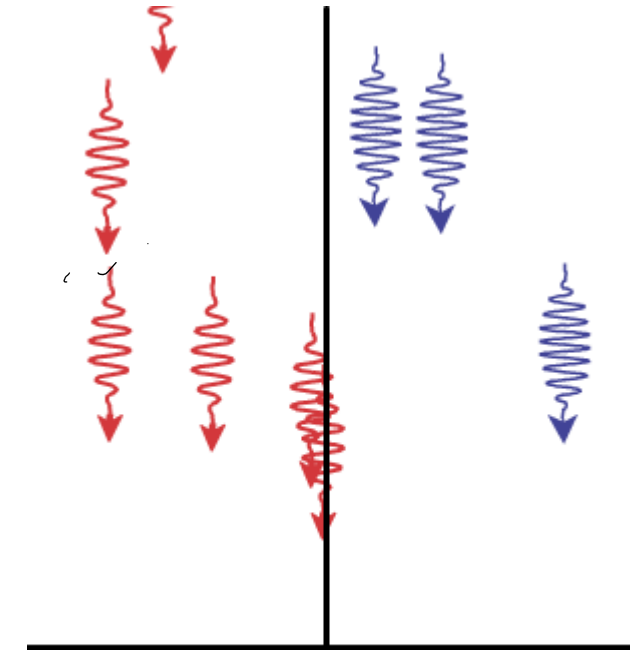
- Now we know how much energy contains in a photon with a particular wavelength.
- Next, how much energy is the sunlight? The answer to this question requires a definition of some terms. The three terms are power, solar intensity, and energy.
- Power is in watts (W), solar intensity is in watts per unit area (W/m^2). Energy is in kilowatt hours (kWh). For examples; homeowners pay for energy in kilowatt hours. Solar modules and arrays produce power in watts.
- The relevant measurement of solar radiation is solar intensity. On a nice sunny day at noon, the solar intensity is usually around 1 kW/m^2 .
- In order to calculate the solar intensity, first we need to understand [photon flux](#) and [spectral irradiance](#).

Photon Flux

- The photon flux is important in determining the number of electrons which are generated and hence the current produced from a solar cell.
- The photon flux is defined as the number of photons per second per unit area:

$$\phi = \frac{\# \text{ of photons}}{m^2.s} \quad (2.6)$$

- The product of the photon flux and the energy of a single photon at a particular wavelength determines the power density of photons at that particular wavelength.
- Since the photon flux gives the number of photons striking a surface in a given time, multiplying the energy of photons by photon flux gives the total energy striking a surface per unit time, which is equivalent to a power density.
- To determine the power density in unit of W/m², the energy of the photon must be in Joules.



For the same light intensity, blue light requires fewer photons since the energy content of each photon is greater.

Figure 2.6

The equation for power density is

$$H \left(\frac{W}{m^2} \right) = \phi \times \frac{hc}{\lambda}, \text{ using SI units}$$

$$H \left(\frac{W}{m^2} \right) = \phi \times q \frac{1.24}{\lambda}, \text{ for wavelength in } \mu m \quad (2.7)$$

$$H \left(\frac{W}{m^2} \right) = \phi \times qE(\text{eV}), \text{ for energy in eV}$$

where ϕ is the photon flux and q is the value of elementary charge 1.6×10^{-19} C.

For example, a light source contains a single wavelength photon with energy of 2eV and the photon flux is $3 \times 10^{21} \text{ m}^{-2}\text{s}^{-1}$. So, the power density of the light source is 960 W/m².

The implication of the above equations is that the photon flux of high energy (or short wavelength) photons need to give a certain power density will be lower than the photon flux of lower energy (or long wavelength) photons required to give the same radiance power density. (Figure 2.6)

Spectral Irradiance

- The spectral irradiance as a function of photon wavelength (or energy), denoted by S , is the most common way of characterizing a light source. It gives the power density at a particular wavelength. The units of spectral irradiance are in $\text{Wm}^{-2}\mu\text{m}^{-1}$. The Wm^{-2} term is the power density at the wavelength $\lambda(\mu\text{m})$. Therefore, the m^{-2} refers to the surface area of the light emitter and the μm^{-1} refers to the wavelength of interest.
- In the analysis of solar cells, the photon flux is often needed as well as the spectral irradiance. The spectral irradiance can be determined from the photon flux by converting the photon flux at a given wavelength to W/m^2 as shown in the section on Photon Flux. The result is then divided by the given wavelength, as shown in the equation 2.8.

$$S(\lambda) = \phi E \frac{1}{\Delta\lambda} \quad (2.8)$$

where in SI units:

$S(\lambda)$ is the spectral irradiance in $\text{Wm}^{-2}\mu\text{m}^{-1}$;

ϕ is the photon flux in # photons $\text{m}^{-2}\text{s}^{-1}$;

E and λ are the energy and wavelength of the photon in joules and meters respectively

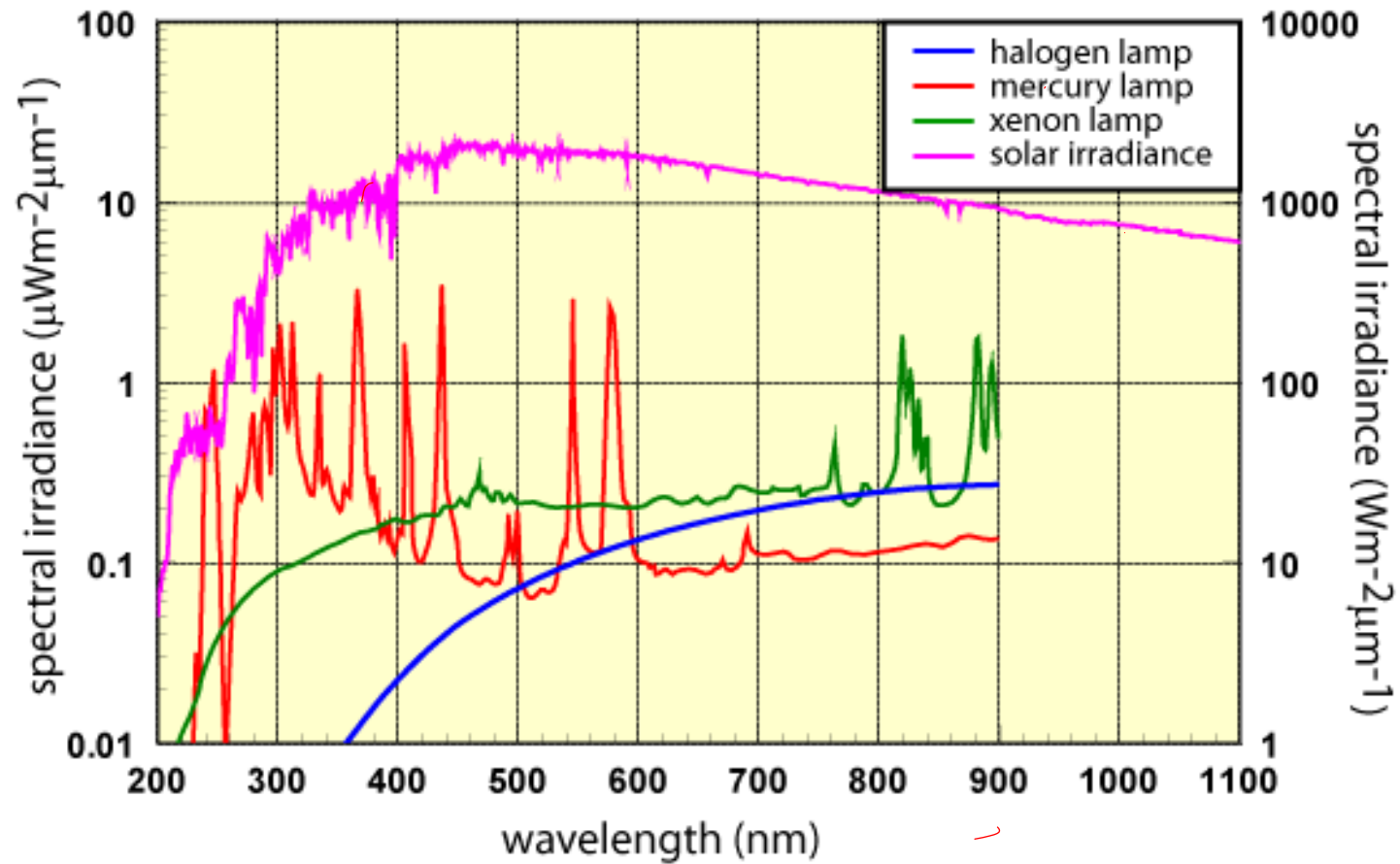


Figure 2.7 The spectral irradiance of artificial light sources (left axis) compared to the spectral irradiance from the sun (right axis).

Radiant Power Density

The total power density emitted from a light source can be calculated by integrating the spectral irradiance over all wavelengths or energies.

$$H = \int_0^{\infty} S(\lambda) d\lambda \quad (2.9)$$

where:

H is the total power density emitted from the light source in W m^{-2} ;
 $S(\lambda)$ is the spectral irradiance in units of $\text{Wm}^{-2}\mu\text{m}^{-1}$; and
 λ is the wavelength.

This radiant power density emitted from the sun is also called solar intensity or (integrated) solar irradiance.

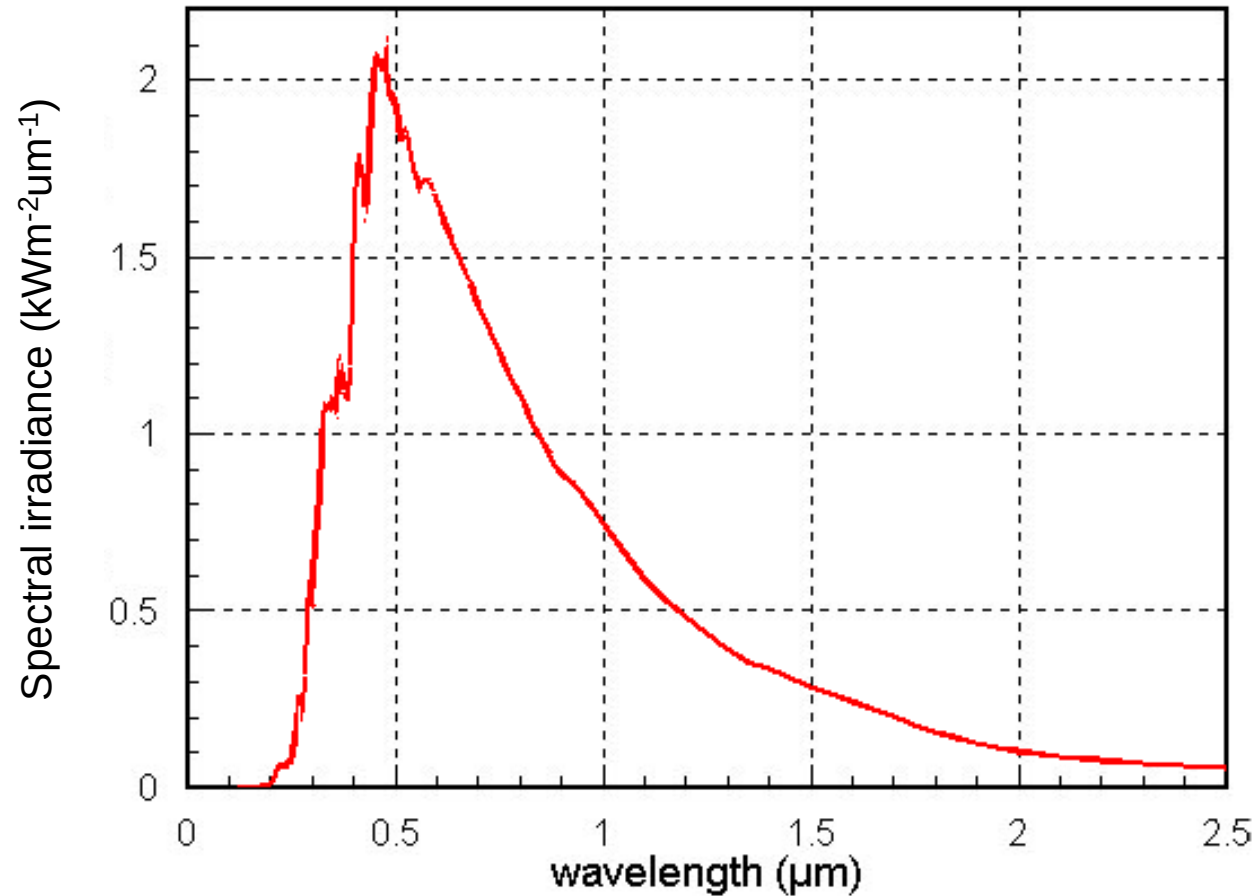


Figure 2.8

Figure 2.8 shows the solar irradiance at the top surface of the earth's atmosphere, which is usually called AM0 spectrum. By integrating the whole range of spectrum, the solar intensity at the top of the earth's atmosphere ($\sim 1366 \text{ W/m}^2$) is obtained.

Black Body Radiation

Many commonly encountered light sources, including the sun and incandescent light bulbs, are closely modelled as "blackbody" emitters. A blackbody absorbs all radiation incident on its surface and emits radiation based on its temperature. Blackbodies derive their name from the fact that, if they do not emit radiation in the visible range, they appear black due to the complete absorption of all wavelengths. The blackbody sources which are of interest to photovoltaics, emit light in the visible region. The spectral irradiance from a blackbody is given by Planck's radiation law, shown in the following equation:

$$S(\lambda) = \frac{2\pi hc^2}{\lambda^5 \left(\exp\left(\frac{hc}{k\lambda T}\right) - 1 \right)} \quad (2.10)$$

where:

λ is the wavelength of light;

T is the temperature of the blackbody (K);

S is the spectral irradiance in $\text{Wm}^{-2}\mu\text{m}^{-1}$; and

h, c and k are Planck's constant (J/s), velocity of light (m/s) and Boltzmann constant (J/K)

If all the parameters are used in SI unit, spectral radiance (S) will result in Wm^{-3} . Dividing by 10^6 gives the conventional units of spectral irradiance in $\text{Wm}^{-2}\mu\text{m}^{-1}$. The notation of $S(\lambda)$ denotes that the spectral irradiance changes with wavelength.

The total power density from a blackbody is determined by integrating the spectral irradiance over all wavelengths which gives:

$$H = \sigma T^4 \quad (2.11)$$

where σ is the Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$) and T is the temperature of the blackbody in kelvin.

An additional important parameter of a blackbody source is the wavelength where the spectral irradiance is the highest, or, in other words the wavelength where most of the power is emitted. The peak wavelength of the spectral irradiance is determined by differentiating the spectral irradiance and solving the derivative when it equals 0. The result is known as Wien's Law and is shown in the following equation:

$$\lambda_p (\mu\text{m}) = \frac{2900}{T} \quad (2.12)$$

where λ_p is the wavelength where the peak spectral irradiance is emitted and T is the temperature of the blackbody (K).

Quiz: What is the peak wavelength of the spectral irradiance emitted from a black body with temperature 6000K and radiance power density?

Ans: 0.483 μm , $7.35 \times 10^7 \text{ W/m}^2$

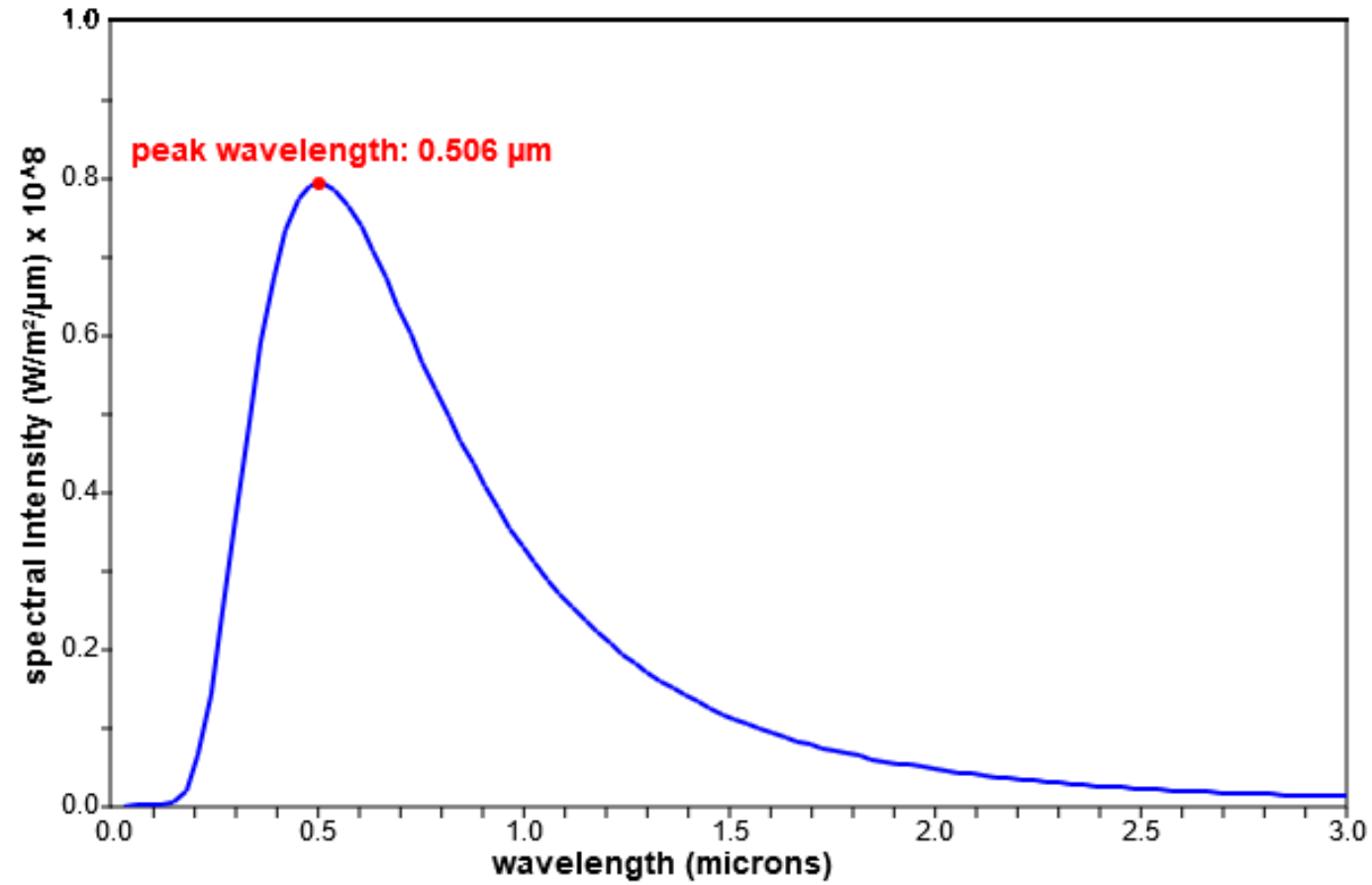


Figure 2.9. The spectral irradiance emitted from a black body with temperature 5730K.

The equations 2.11 and 2.12 show that as the temperature of a blackbody increases, the spectral distribution and power of light emitted change. For example, near room temperature, a blackbody emitter (such as a human body or light bulb which is turned off) will emit low power radiation at wavelengths predominantly greater than $1\mu\text{m}$, well outside the visual range of human observation. If the blackbody is heated to 3000 K, it will glow red because the spectrum of emitted light shifts to higher energies and into the visible spectrum. If the temperature of the filament is further increased to 6000 K, radiation is emitted at wavelengths across the visible spectrum from red to violet and the light appears white. The graphs below compare the spectral irradiance of a blackbody at these three temperatures. The room temperature case of 300K (the black dotted line) has essentially no power emitted in the visible and near infrared portions of the spectrum shown on the graph. Because of the huge variation in both emitted power and the range of wavelengths over which the power is emitted, the log graph below demonstrates more clearly the variation in the emitted blackbody spectrum as a function of temperature.

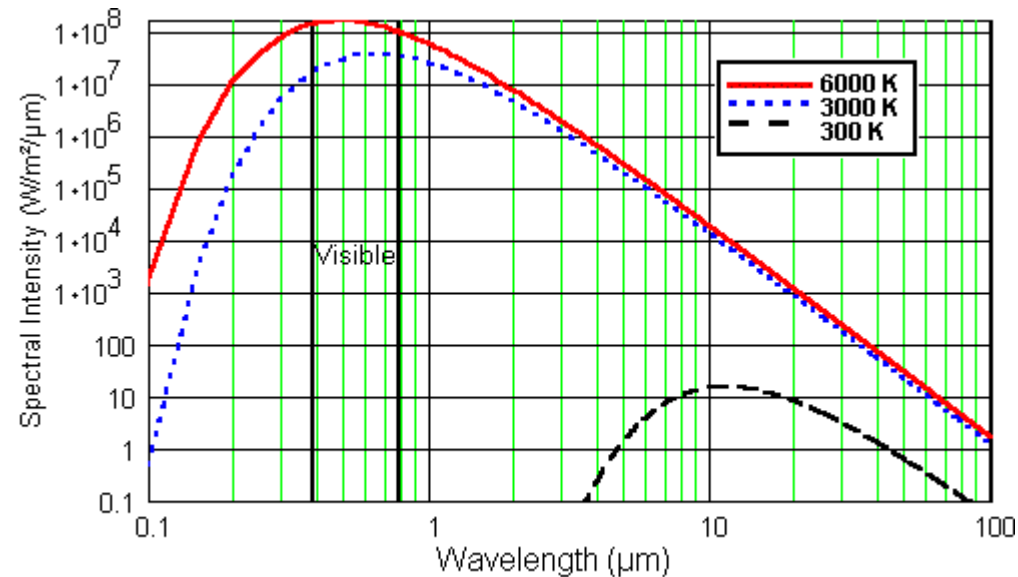


Figure 2.10

The Sun

The sun is a hot sphere of gas whose internal temperatures reach over 20 million degrees kelvin due to nuclear fusion reactions at the sun's core which convert hydrogen to helium. The radiation from the inner core is not visible since it is strongly absorbed by a layer of hydrogen atoms closer to the sun's surface. Heat is transferred through this layer by convection.

The surface of the sun, called the photosphere, is at a temperature of about 6000K and closely approximates a blackbody. For simplicity, the 6000 K spectrum is commonly used in detailed balance calculations but temperatures of 5762 ± 50 K and 5730 ± 90 have also been proposed as a more accurate fit to the sun's spectrum. Figure 2.9 is the spectral irradiance emitted from the sun.

The total power emitted by the sun is calculated by multiplying the emitted power density by the surface area of the sun which gives 9.5×10^{25} W.

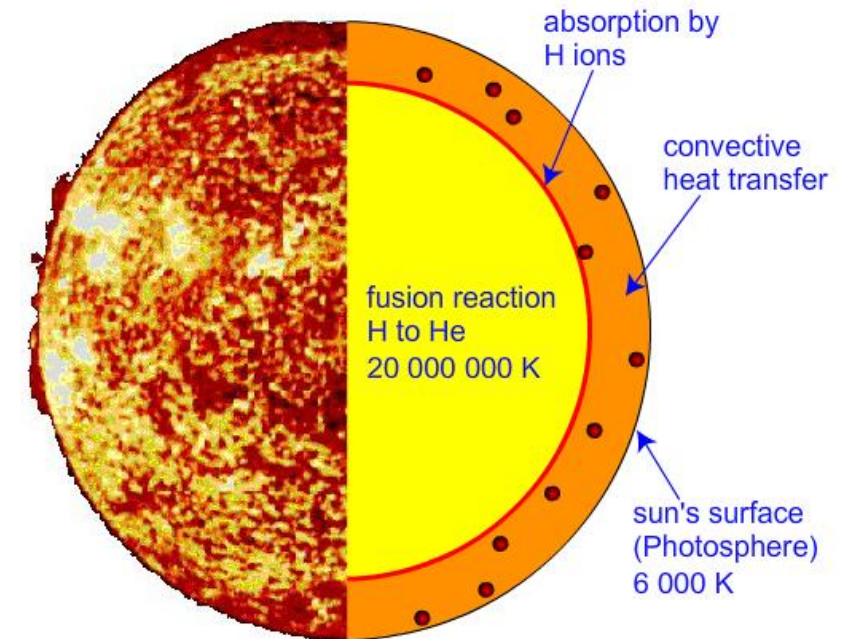


Figure 2.11

Solar Radiation in Space

Only a fraction of the total power emitted by the sun impinges on an object in space which is some distance from the sun. The solar irradiance (H_0 in W/m^2) is the power density incident on an object due to illumination from the sun. At the sun's surface, the power density is that of a blackbody at about 6000K and the total power from the sun is this value multiplied by the sun's surface area. However, at some distance from the sun, the total power from the sun is now spread out over a much larger surface area and therefore the solar irradiance on an object in space decreases as the object moves further away from the sun.

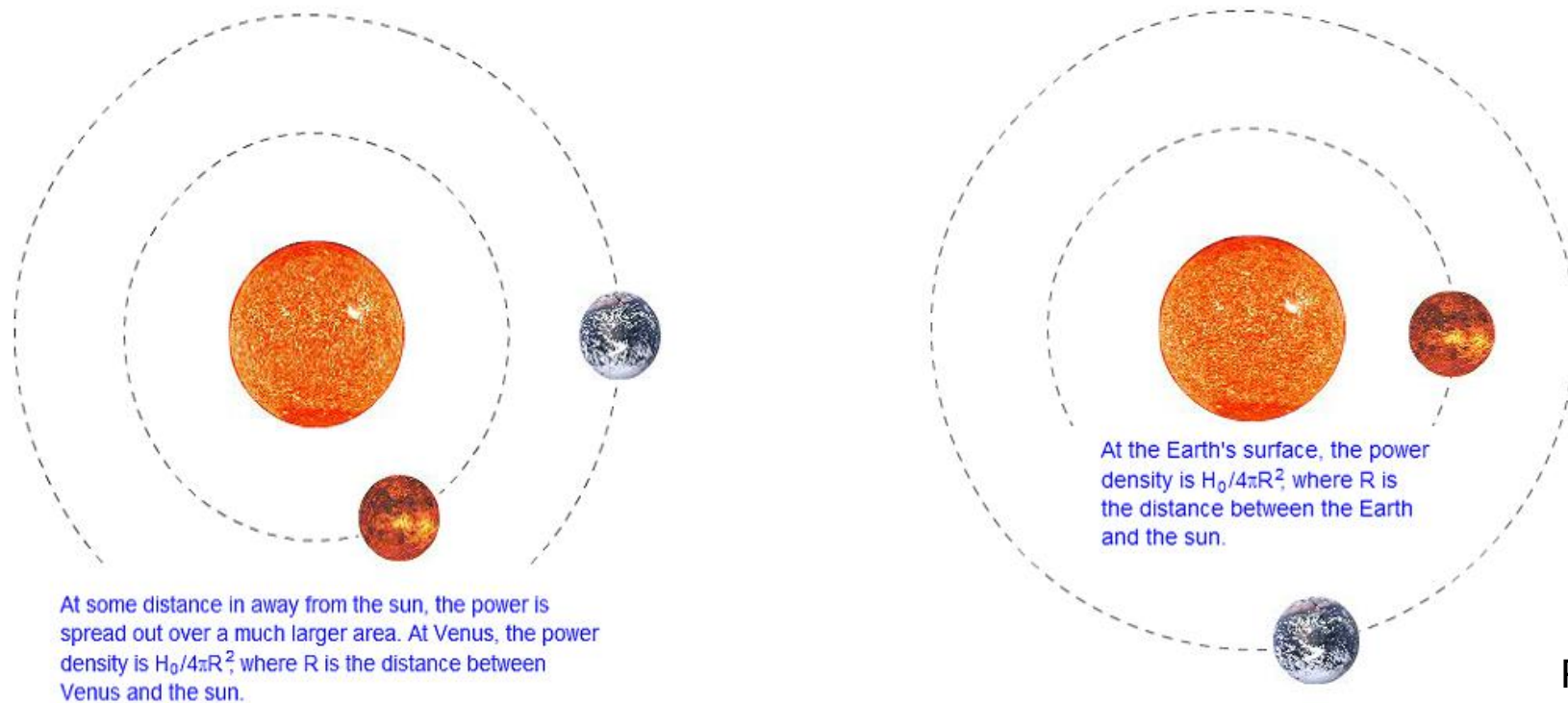


Figure 2.12

The solar irradiance on an object some distance D from the sun is found by dividing the total power emitted from the sun by the surface area over which the sunlight falls. The total solar radiation emitted by the sun is given by σT^4 multiplied by the surface area of the sun ($4\pi R_{\text{sun}}^2$) where R_{sun} is the radius of the sun. The surface area over which the power from the sun falls will be $4\pi D^2$. Where D is the distance of the object from the sun. Therefore, the solar radiation intensity, H_0 in (W/m^2), incident on an object is:

$$H_0 = \frac{R_{\text{sun}}^2}{D^2} H_{\text{sun}} \quad (2.13)$$

where:

H_{sun} is the power density at the sun's surface (in W/m^2) as determined by Stefan-Boltzmann's blackbody equation;

R_{sun} is the radius of the sun in meters as shown in the figure below; and

D is the distance from the sun in meters as shown in the figure below.

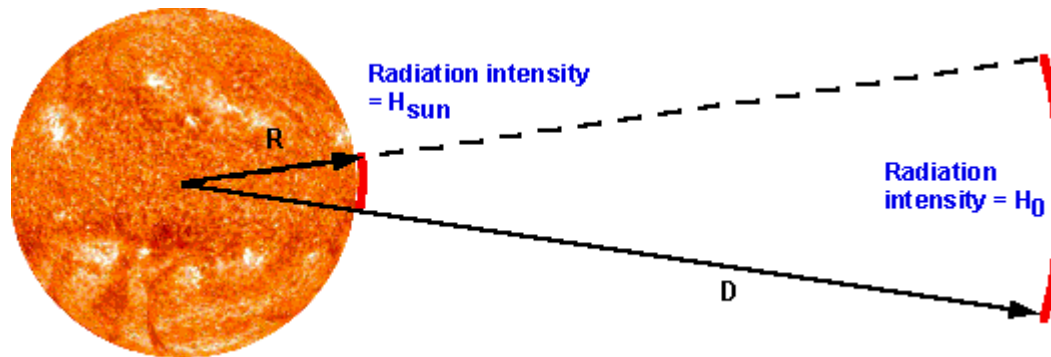


Figure 2.13

The table below gives standardized values for the radiation at each of the planets but by entering the distance you can obtain an approximation.

Planet	Distance (x 10 ⁹ m)	Mean Solar Irradiance (W/m ²)
Mercury	57	9116.4
Venus	108	2611.0
Earth	150	1366.1
Mars	227	588.6
Jupiter	778	50.5
Saturn	1426	15.04
Uranus	2868	3.72
Neptune	4497	1.51
Pluto	5806	0.878

Solar Radiation Outside the Earth's Atmosphere

The solar radiation outside the earth's atmosphere is calculated using the radiant power density (H_{sun}) at the sun's surface ($5.961 \times 10^7 \text{ W/m}^2$), the radius of the sun (R_{sun}), and the distance between the earth and the sun. The calculated solar irradiance at the Earth's atmosphere is about 1.36 kW/m^2 . The geometrical constants used in the calculation of the solar irradiance incident on the Earth are shown in Figure 2.14 .

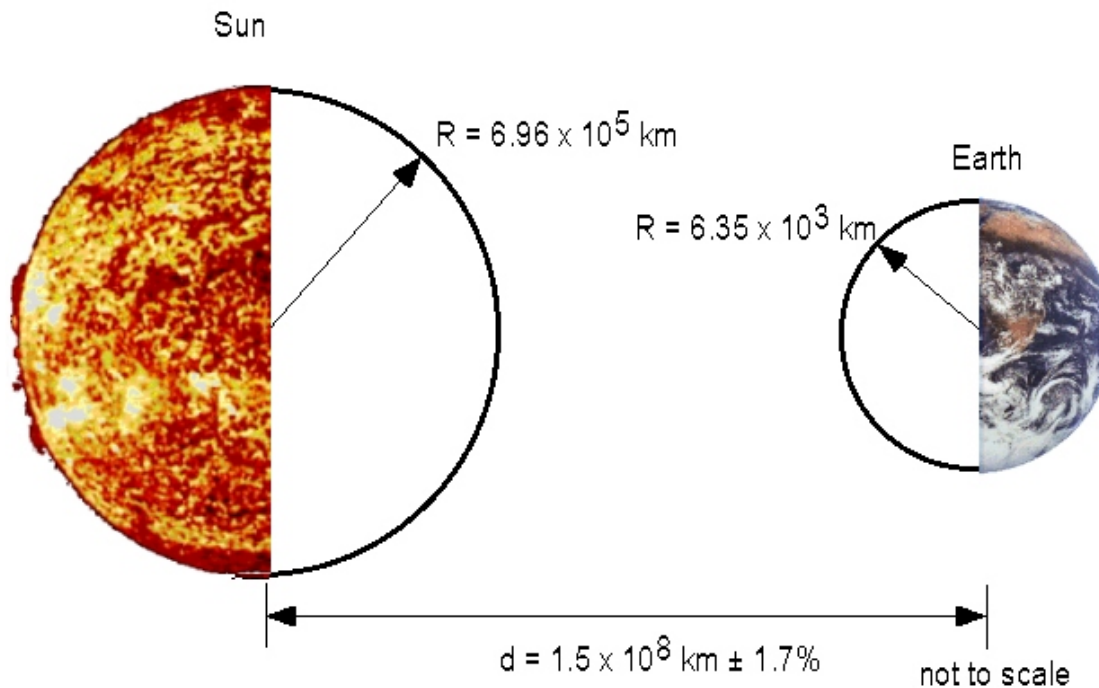


Figure 2.14. Geometrical constants for finding the Earth's solar irradiance. The diameter of the Earth is not needed but is included for the sake of completeness.

The actual power density varies slightly since the Earth-Sun distance changes as the Earth moves in its elliptical orbit around the sun, and because the sun's emitted power is not constant. The power variation due to the elliptical orbit is about 3.4%, with the largest solar irradiance in January and the smallest solar irradiance in July. An equation which describes the variation through out the year just outside the earth's atmosphere is:

$$\frac{H}{H_{constant}} = 1 + 0.033 \cos \left(\frac{360(n-2)}{365} \right) \quad (2.14)$$

where:

H is the radiant power density outside the Earth's atmosphere (in W/m²);

H_{constant} is the value of the solar constant, 1.366 kW/m²; and n is the day of the year.

These variations are typically small and for photovoltaic applications the solar irradiance can be considered constant. The value of the solar constant and its spectrum have been defined as a standard value called air mass zero (AM0) and takes a value of 1.366 kW/m².

Solar Radiation at the Earth's Surface

While the solar radiation incident on the Earth's atmosphere is relatively constant, the radiation at the Earth's surface varies widely due to:

- atmospheric effects, including absorption and scattering;
- local variations in the atmosphere, such as water vapour, clouds, and pollution;
- latitude of the location; and
- the season of the year and the time of day.

The above effects have several impacts on the solar radiation received at the Earth's surface. These changes include variations in the overall power received, the spectral content of the light and the angle from which light is incident on a surface. In addition, a key change is that the variability of the solar radiation at a particular location increases dramatically. The variability is due to both local effects such as clouds and seasonal variations, as well as other effects such as the length of the day at a particular latitude. Desert regions tend to have lower variations due to local atmospheric phenomena such as clouds. Equatorial regions have low variability between seasons.

The amount of energy reaching the surface of the Earth every hour is greater than the amount of energy used by the Earth's population over an entire year.

Atmospheric Effects

Atmospheric effects have several impacts on the solar radiation at the Earth's surface. The major effects for photovoltaic applications are:

- a reduction in the power of the solar radiation due to absorption, scattering and reflection in the atmosphere;
- a change in the spectral content of the solar radiation due to greater absorption or scattering of some wavelengths;
- the introduction of a diffuse or indirect component into the solar radiation; and
- local variations in the atmosphere (such as water vapor, clouds and pollution) which have additional effects on the incident power, spectrum and directionality.

These effects are summarized in Figure 2.15.

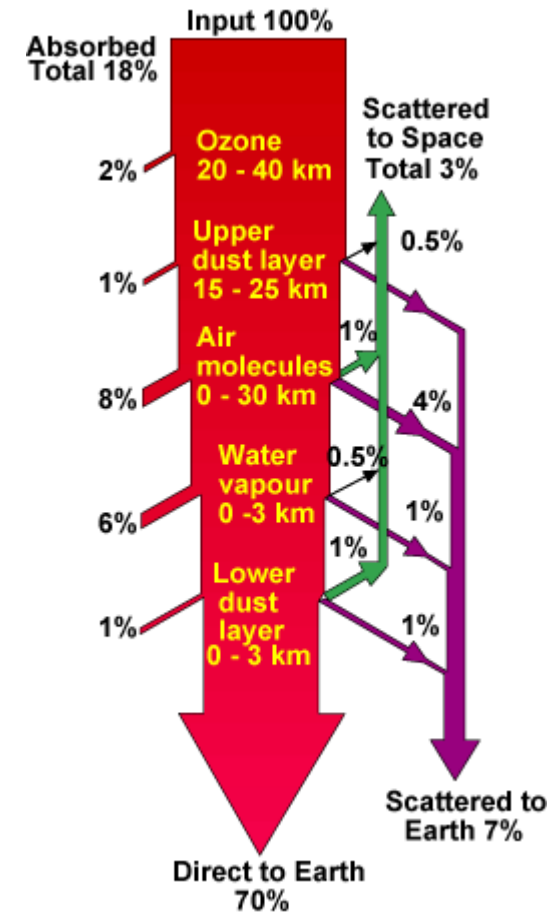


Figure 2.15. Typical clear sky absorption and scattering of incident sunlight

Absorption in the Atmosphere

- As solar radiation passes through the atmosphere, gasses, dust and aerosols absorb the incident photons. Specific gasses, notably ozone (O_3), carbon dioxide (CO_2), and water vapor (H_2O), have very high absorption of photons that have energies close to the bond energies of these atmospheric gases. This absorption yields **deep troughs** in the spectral radiation curve. For example, much of the far infrared light **above $2\ \mu m$** is absorbed by **water vapor and carbon dioxide**. Similarly, most of the ultraviolet light **below $0.3\ \mu m$** is absorbed by **ozone** (but not enough to completely prevent sunburn!).
- While the absorption by specific gasses in the atmosphere change the spectral content of the terrestrial solar radiation, they have a relatively minor impact on the overall power. Instead, the major factor reducing the power from solar radiation is the absorption and scattering of light due to **air molecules and dust**. This absorption process **does not produce the deep troughs** in the spectral irradiance, but rather causes a **power reduction** dependent on the path length through the atmosphere. When the sun is overhead, the absorption due to these atmospheric elements causes a relatively uniform reduction across the visible spectrum, so the incident light appears white. However, for longer path lengths, higher energy (lower wavelength) light is more effectively absorbed and scattered. Hence in the morning and evening the sun appears much redder and has a lower intensity than in the middle of the day.

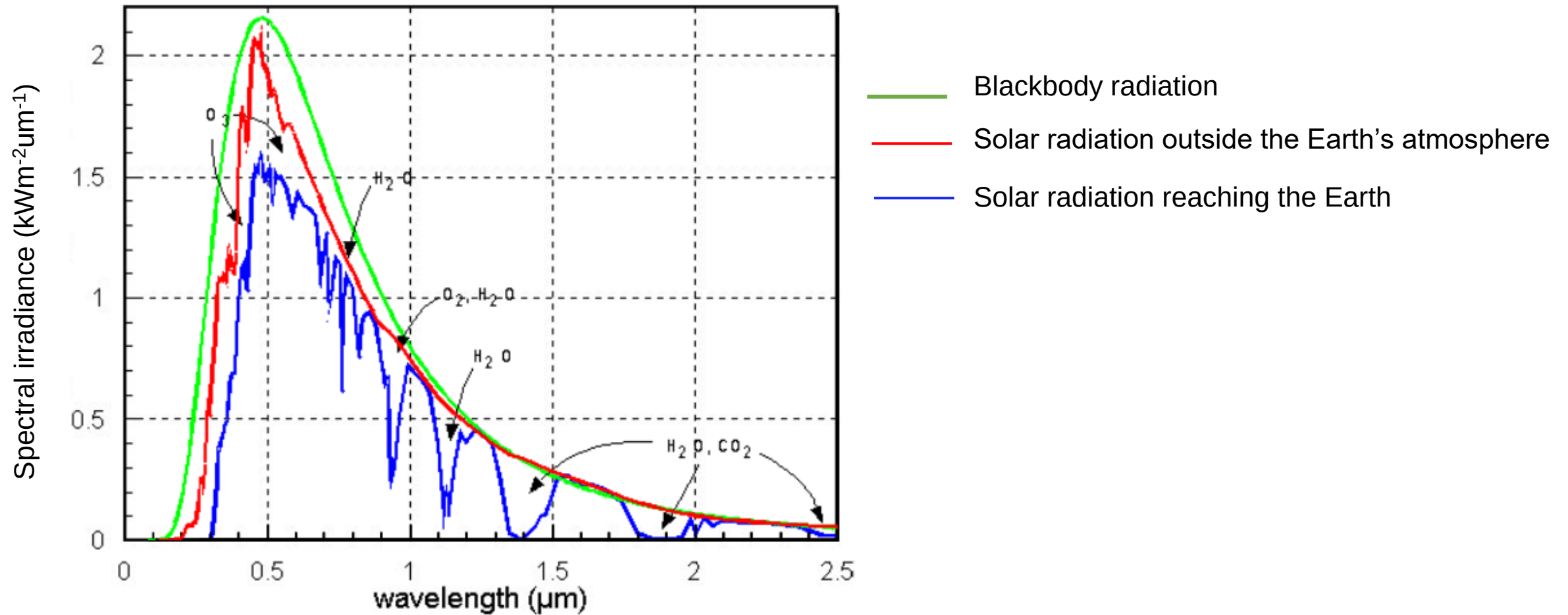


Figure 2.16. A comparison of solar radiation outside the Earth's atmosphere with the amount of solar radiation reaching the Earth itself.

Direct and Diffuse Radiation Due to Scattering of Incident Light

- Light is absorbed as it passes through the atmosphere and at the same time it is subject to scattering. One of the mechanisms for light scattering in the atmosphere is known as Rayleigh scattering which is caused by molecules in the atmosphere. Rayleigh scattering is particularly effective for short wavelength light (that is blue light) since it has a λ^{-4} dependence. In addition to Rayleigh scattering, aerosols and dust particles contribute to the scattering of incident light known as Mie scattering.
- Scattered light is undirected, and so it appears to be coming from any region of the sky. This light is called "diffuse" light. Since diffuse light is primarily "blue" light, the light that comes from regions of the sky other than where the sun is, appears blue. In the absence of scattering in the atmosphere, the sky would appear black, and the sun would appear as a disk light source. On a clear day, about 10% of the total incident solar radiation is diffuse.

Air Mass

On passing through the atmosphere, the solar spectrum is partially attenuated by the absorption of oxygen, ozone in the ultraviolet region and water vapor, carbon dioxide, methane in the infrared. This attenuation is described by the *Air Mass* (AM) factor since the absorption increases with the mass of air through which the radiation pass.

For a thickness of l_0 of the atmosphere, the path length l through the atmosphere for radiation at an incident angle α relative to the normal to the earth's surface is given by

$$l_0 = l \cos \alpha \quad (2.15)$$

The ratio of l to l_0 is called AM factor. Mathematically,

$$AM = \frac{l}{l_0} = \frac{1}{\cos \alpha} \quad (2.16)$$

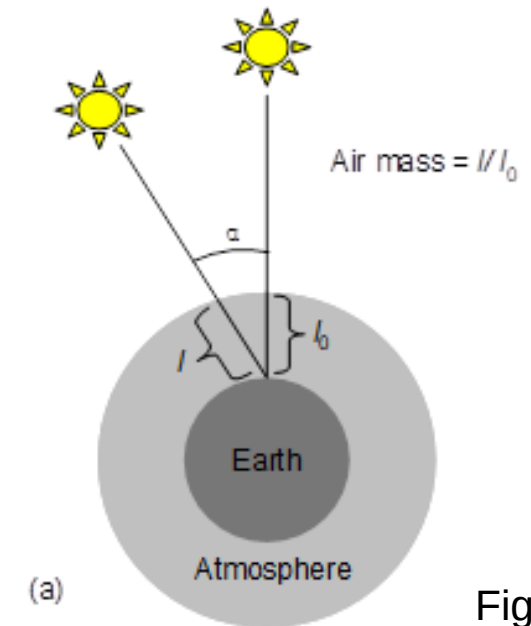


Figure 2.17

- The spectrum outside the atmosphere is denoted as AM0 and that on the surface of the earth for perpendicular incidence (when the sun is directly overhead) as AM1.
- AM0 spectrum is usually called ASTM E-490, which is used for aerospace community and has an integrated spectral irradiance of 1366 W/m^2 .
- The photovoltaic community developed another spectrum called AM1.5, which corresponds to a solar incident angle of 48° relative to the surface normal (or 42° above horizon), for terrestrial applications. This AM1.5 is called ASTM G-173.
- The AM1.5 spectra are further divided into *global* and *direct* terrestrial reference spectra, notated as AM1.5D and AM1.5G, respectively. AM1.5G includes both direct and diffuse radiation whereas AM1.5D includes direct radiation only.
- The AM1.5 Global spectrum is designed for flat plate modules and has an integrated power of 1000 W/m^2 (100 mW/cm^2).
- The AM1.5 Direct (+circumsolar) spectrum is defined for solar concentrator work. It includes the direct beam from the sun plus the circumsolar component in a disk 2.5 degrees around the sun. The direct plus circumsolar spectrum has an integrated power density of 900 W/m^2 .

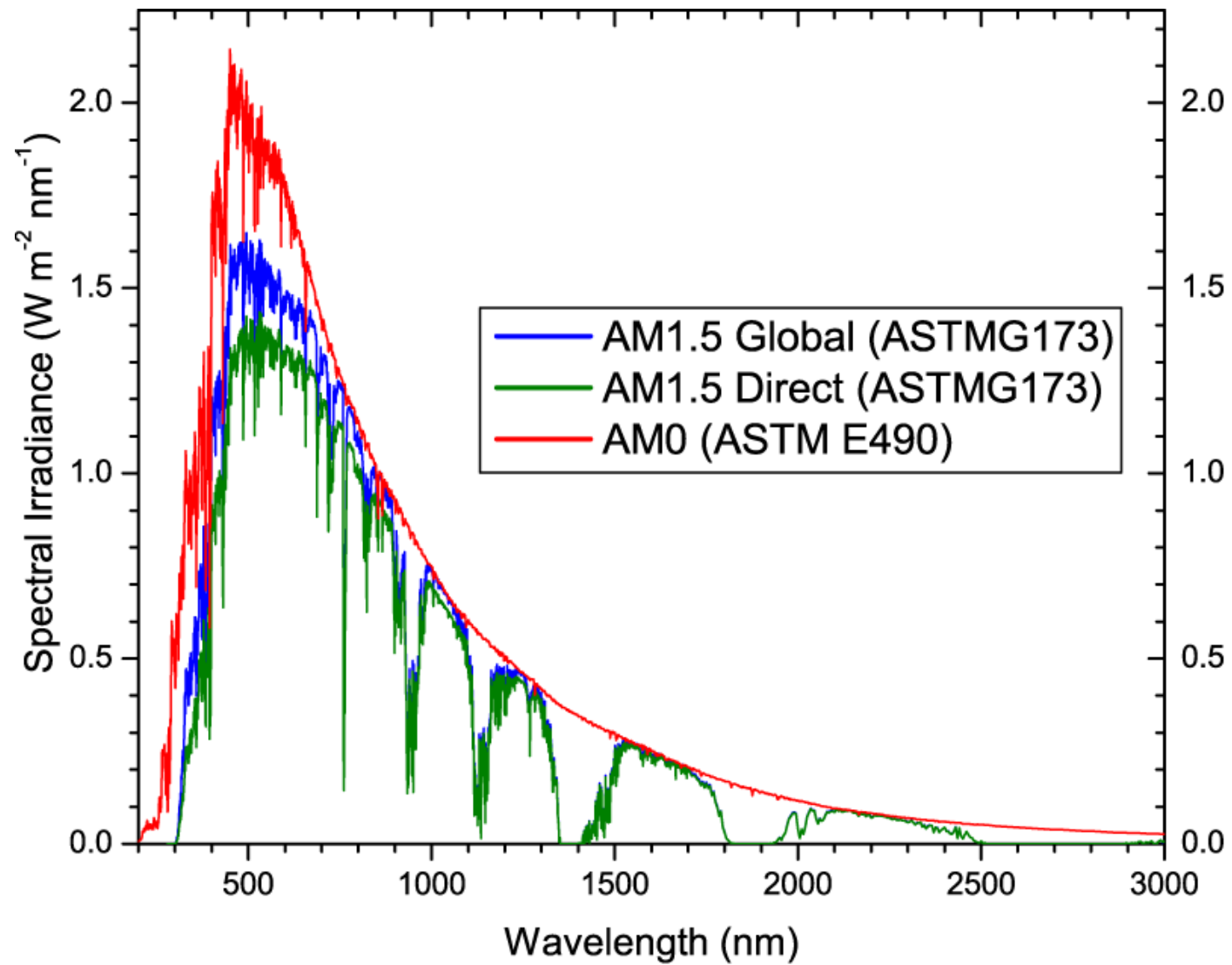


Figure 2.18. Standard Solar Spectra for space and terrestrial use.

Motion of the Sun

The apparent motion of the sun, caused by the rotation of the Earth about its axis, changes the angle at which the direct component of light will strike the Earth. From a fixed location on Earth, the sun appears to move throughout the sky. The position of the sun depends on the location of a point on Earth, the time of day and the time of year.

This apparent motion of the sun has a major impact on the amount of power received by a solar collector. When the sun's rays are perpendicular to the absorbing surface, the power density on the surface is equal to the incident power density. However, as the angle between the sun and the absorbing surface changes, the intensity on the surface is reduced. When the module is parallel to the sun's rays (and the angle to the module normal = 90°) the intensity of light essentially falls to zero. For intermediate angles, the relative power density is $\cos(\theta)$ where θ is the angle between the sun's rays and the module normal.

The angle between the sun and a fixed location on Earth depends on the particular location (the longitude of the location), the time of year and the time of day. In addition, the time at which the sun rises and sets depends on the longitude of the location. Therefore, complete modeling of the sun's angle to a fixed position on Earth requires the latitude, longitude, day of the year, and time of day.

Sun Position Calculator

Solar Radiation on a tilted surface

Calculation of Solar Insolation

Refer to Section 2.4 of www.pveducation.org.